Classification of Algebraic Varieties

Christopher Hacon

University of Utah

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Outline of the talk

1. Introduction

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2. Irreducible subsets

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3. Curves

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5. Higher dimensions

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Polynomial equations

Algebraic Geometry is concerned with the study of solutions of polynomial equations say n-equations in m-variables

$$\begin{cases} P_1(x_1,\ldots,x_m)=0 \\ \qquad \ldots \\ \qquad \ldots \\ P_n(x_1,\ldots,x_m)=0. \end{cases}$$

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$$P(x) = a_d x^d + a_{d-1} x^{d-1} + \ldots + a_1 x + a_0 \qquad a_d \neq 0$$

(a polynomial of degree d in $\mathbb{C}[x]$).

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(a polynomial of degree d in $\mathbb{C}[x]$).
- If $a_i \in \mathbb{C}$, then by the Fundamental Theorem of Algebra, we may find $c_i \in \mathbb{C}$ such that $P(x)$ factors as
$$P(x) = a_d(x - c_1)(x - c_2)\cdots(x - c_d).$$

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- Therefore a polynomial of degree d always has exactly d solutions or roots (when counted with multiplicity).
- If one is interested in solutions that belong to $\mathbb{R}$ (or $\mathbb{Q}$, or $\mathbb{Z}$ etc.), then the problem is much more complicated, but at least we know that there are at most d solutions.

Complex solutions

Throughout this talk I will always look for complex solutions $(z_1, \ldots, z_m) \in \mathbb{C}^m$ to polynomial equations

$$P_1(x_1, \ldots, x_m) = 0 \qquad \ldots \qquad P_n(x_1, \ldots, x_m) = 0$$

where $P_i \in \mathbb{C}[x_1, \ldots, x_m]$.

Projective space

- It is more convenient to work on a compactification $\mathbb{C}^N \subset \mathbb{P}^N_{\mathbb{C}}$ where $\mathbb{P}^N_{\mathbb{C}} = \{\mathbb{C}^{N+1} - (0, \ldots, 0)\}/\{\mathbb{C}^*\}$ is projective N-space.

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- We have that $\mathbb{P}^0_{\mathbb{C}}$ is a point, $\mathbb{P}^1_{\mathbb{C}} \cong S^2$ is the Riemann sphere and $\mathbb{P}^N_{\mathbb{C}} = \mathbb{C}^N \cup \mathbb{P}^{N-1}_{\mathbb{C}}$.

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- We then replace $P(x_1,\dots,x_N)$ by the coresponding homogeneous polynomial in $\mathbb{C}[x_0,\dots,x_N]$. Eg. $x_1^2 + x_2^3 \Rightarrow x_0x_1^2 + x_2^3$.

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- We then replace $P(x_1, \dots, x_N)$ by the coresponding homogeneous polynomial in $\mathbb{C}[x_0, \dots, x_N]$. Eg. $x_1^2 + x_2^3 \Rightarrow x_0 x_1^2 + x_2^3$.
- Note that if $P(x_1, \dots, x_N)$ is homogeneous and $\lambda \in \mathbb{C}^*$, then $P(c_1, \dots, c_N) = 0$ iff $P(\lambda c_1, \dots, \lambda c_N) = 0$. Thus the zeroes of $P(x_1, \dots, x_N)$ are a well defined subset of $\mathbb{P}^N_{\mathbb{C}}$.

Bezout's Theorem

Theorem (Bezout's Theorem)

Let $P_1(x_0,\dots,x_n),\dots,P_n(x_0,\dots,x_n)$ be n homogeneous polynomials of degrees $d_1,\dots,d_n$. If the set

$$\bigcap_{i=1}^{n}\{P_i(x_0,\dots,x_n)=0\}\subset\mathbb{P}^n_{\mathbb{C}}$$

is finite, then it consist of exactly $d_1\cdots d_n$ points (counted with multiplicity).

Eg. two distinct lines in $\mathbb{P}^2_{\mathbb{C}}$ always meet in one point.

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Irreducible subsets

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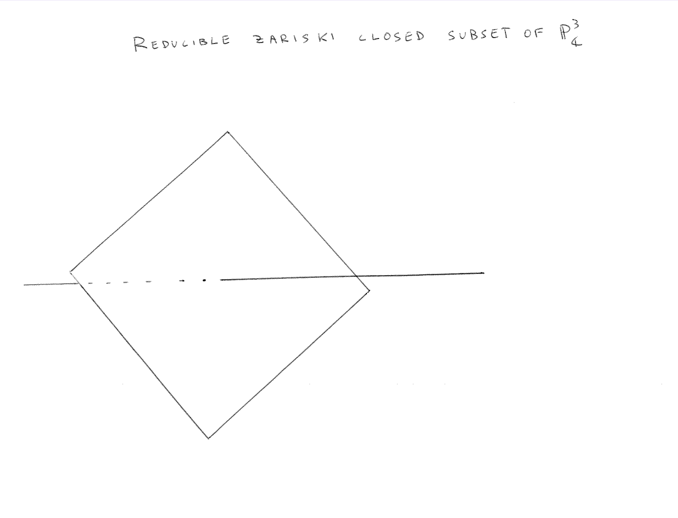
- If $P(x, y, z) \in \mathbb{C}[x, y, z]$ is homogeneous of degree $r > 0$, then
$$X = \{P(x, y, z) = 0\} \subset \mathbb{P}^2_{\mathbb{C}}$$
is irreducible if and only if $P(x, y, z)$ does not factor.

Reducible set



Varieties

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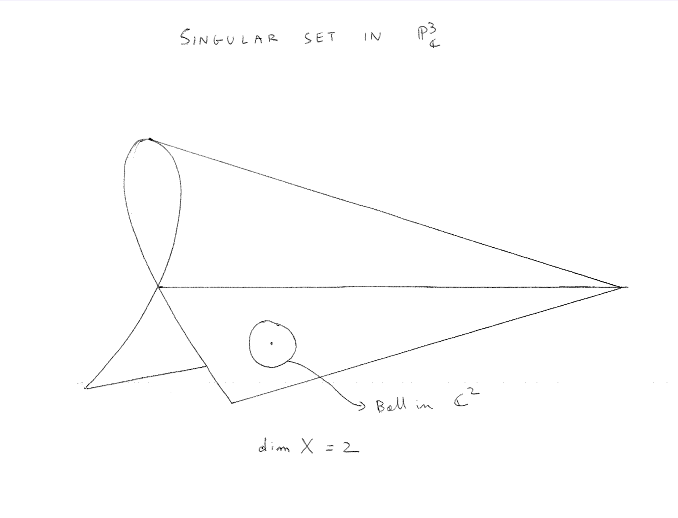
- We then say that x is a **smooth** point and that X has **dimension** d.
- Most Zariski closed sets are irreducible and smooth.
- By a deep result of Hironaka, there exists a **desingularization** of X, that is a morphism $f : Y \to X$, such that Y is everywhere smooth and f is the identity over any smooth point of X.

Varieties



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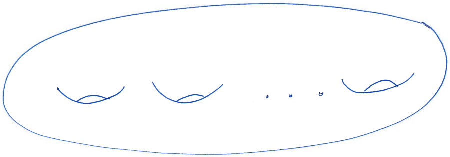
- The simplest curves are **rational curves** i.e. curves of genus 0.
- There is only one such curve $X = \mathbb{P}^1_{\mathbb{C}}$.
- When $g = 1$ we have a 1-parameter family of **elliptic curves** $X_t = \{y^2 - x(x-1)(x-t) = 0\}$.

Curves

$$\mathbb{P}^2_{\mathbb{C}} \supset \{Y^2Z^{d-2} = X(X-Z)(X-2Z)\cdots(X-(d-1)Z)\}$$

d even



genus $= \frac{d}{2} - 1$

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$$\bigcap_{i=1}^{n}\{P_i(x_1,\ldots,x_m)=0\}\subset \mathbb{P}^m_{\mathbb{C}}.$$

- We would like to find a natural (canonical) description for $X \subset \mathbb{P}^m_{\mathbb{C}}$.

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- If L is a line bundle on X and $s_0, \ldots, s_m$ is a basis of $H^0(X, L)$, then there is a map $\phi_L : X \dashrightarrow \mathbb{P}^m_{\mathbb{C}}$ given by

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  (undefined at the common zeroes of the s_i).
- To define an embedding of X in $\mathbb{P}^m_{\mathbb{C}}$, we must find a **very ample** line bundle L (i.e. ϕ_L defines an embedding or equivalently the sections of L distinguish different points and tangent directions).

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Theorem

Let X be a curve of genus $g \geq 2$, then for any $k \geq 3$ the sections of $H^0(X, \omega_X^{\otimes k})$ define an embedding

$$\phi_{\omega_X^{\otimes k}} : X \hookrightarrow \mathbb{P}_{\mathbb{C}}^{(2k-1)(g-1)-1}.$$

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- Thus X is defined by at most $(3(2g-2))^2$ equations of degree at most $3(2g-2)$.
- Hence these curves depend on finitely many parameters (in fact on only $3g-3$ parameters).

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- In coordinates this may be described as follows. If $r_0, \cdots, r_m$ generate $R(\omega_X)$ and $K = \ker\left(\mathbb{C}[x_0, \ldots, x_m] \to R(\omega_X)\right)$, then $X \subset \mathbb{P}^m_{\mathbb{C}}$ is defined by the equations in K.

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The canonical ring (in dimension $d \geq 2$)

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- We have $\kappa(X) = \max_{k>0}\{\dim \phi_{\omega_X^{\otimes k}}(X)\}$ so that $\kappa(X) \in \{-1, 0, 1, \cdots, d\}$. (For $d = 1$ we have $g = 0, 1, \geq 2$, iff $\kappa(X) = -1, 0, 1$.)

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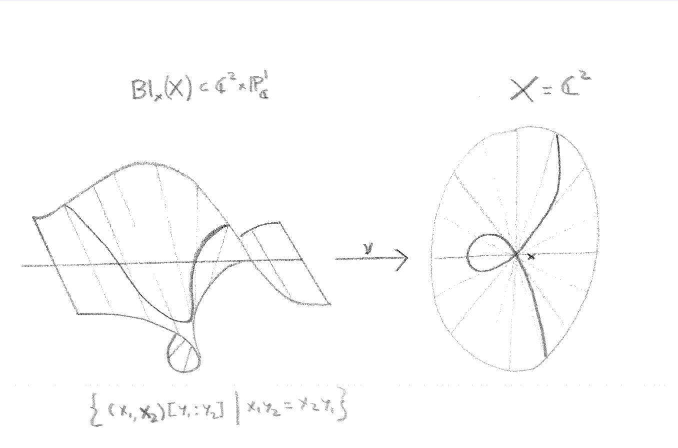
- The morphism ν may be viewed as a surgery that replaces the point x by a rational curve $E \cong \mathbb{P}^1 = \mathbb{P}(T_x X)$.
- The exceptional curve E is called a -1 **curve** since

$$c_1(\omega_{\tilde{X}}) \cdot E = \deg(\omega_{\tilde{X}}|_E) = -1.$$

Blow up


$Bl_x(X) \subset \mathbb{C}^2 \times \mathbb{P}^1_{\mathbb{C}}$
$X = \mathbb{C}^2$
ν
x
$\{(x_1, x_2)[y_1 : y_2] \mid x_1 y_2 = x_2 y_1\}$

Birational geometry of surfaces II

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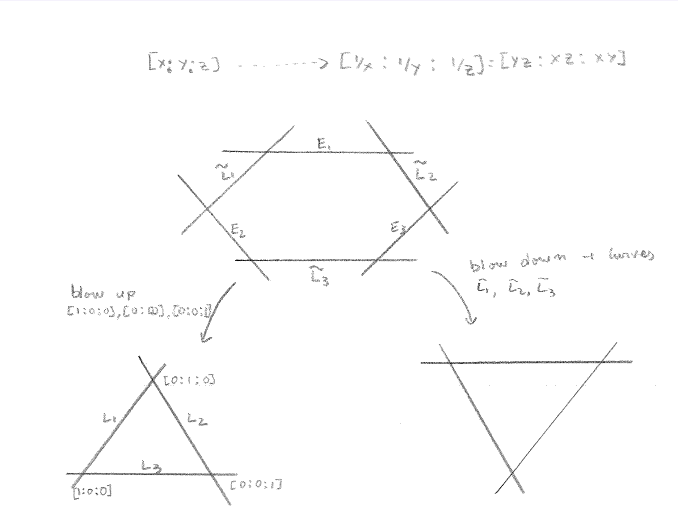
- It is then natural to study surfaces up to birational equivalence.

Birational map



Birational geometry of surfaces III

- By a Theorem of Castelnuovo, one may reverse the blowing up procedure i.e. given any surface X and any -1 curve $E \subset X$, there exists a morphism $\nu : X \to X_1$ such that $X = \mathrm{Bl}_x(X_1)$ for some point $x \in X_1$.

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- In fact one can show that $\omega_{X_{\min}}$ is **semiample** (i.e. there is an integer $m>0$ such that the sections of $H^0(\omega_{X_{\min}}^{\otimes m})$ have no common zeroes).

Birational geometry of surfaces V

- Therefore, $\phi_{\omega_{X_{\min}}^{\otimes m}} : X_{\min} \to \mathbb{P}^N$ is a morphism, and $\omega_{X_{\min}}^{\otimes m} \cong \phi^* \mathcal{O}_{\mathbb{P}^N}(1)$.

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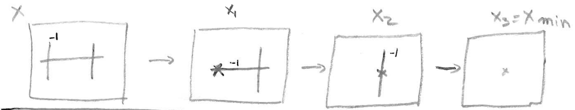
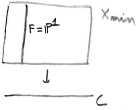
- Conclusion: If $\kappa(X) = 2$, then we know that if $c_1(\omega_{X_{\min}})^2 \leq M$ then there are only finitely many families of such surfaces (modulo birational isomorphism).

Review



$\kappa(X) < 0$

eg $\mathbb{P}^1 \times C$

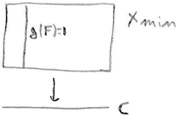


$\kappa(X) = 0 \Rightarrow X_{min}$ is Abelian, K3, Enriques or bielliptic surface

eg $E \times E'$

$g(E) = g(E') = 1$

$K(X) = 1$



eg $F \times C$

$g(F) = 1$

$g(C) \geq 2$

$K(X) = 2$ eg $C \times C'$ $g(C), g(C') \geq 2$

$$\phi_{\omega_{X_{min}}^{\otimes 5}} : X_{min} \overset{\text{birat}}{\dashrightarrow} \bar{X} \subset \mathbb{P}^3$$

$\uparrow$ deg $\leq c_1(\omega_{X_{min}})^2 \cdot 25$

If $c_1(\omega_{X_{min}})^2 \leq M$ then X_{min} belongs to finitely many families

Outline of the talk

1 Introduction

2 Irreducible subsets

3 Curves

4 Surfaces

5 Higher dimensions

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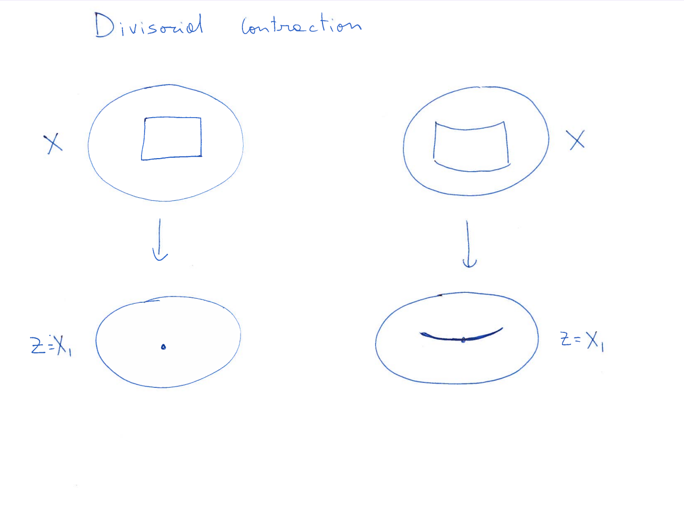
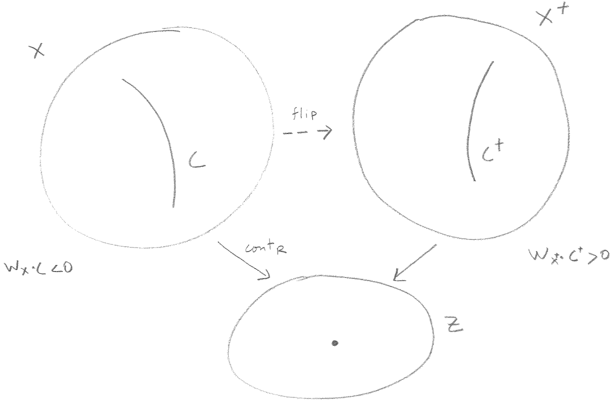
- $X_{\min}$ and X_{can} can be mildly singular.

Divisorial Contraction



Flip



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- Remark: Note that $\rho(X^+) = \rho(X)$. Never-the-less, it is conjectured that there is no infinite sequence of flips.

Higher dimensions

Theorem (Birkar-Cascini-Hacon-M^cKernan)

Let X be a variety of general type (i.e. $\kappa(X) = \dim(X)$), then there exists a finite sequence of flips and divisorial contractions

$$X \dashrightarrow X_1 \dashrightarrow \ldots \dashrightarrow X_k = X_{\min}$$

such that $\omega_{X_{\min}}$ is nef and semiample.

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- The minimal model $X_{\min} = X_k$ has mild singularities but it is not uniquely determined.
- The **canonical model** $X_{\mathrm{can}} = \mathrm{Proj}(R(\omega_X))$ is unique, X_{can} has mild singularities and $\omega_{X_{\mathrm{can}}}^{\otimes m}$ is very ample (some $m > 0$).

Finite generation

Theorem (Birkar-Cascini-Hacon-M^{c}Kernan, Siu)

The canonical ring

$$R(\omega_X)=\bigoplus_{k\geq 0} H^0(X,\omega_X^{\otimes k})$$

is finitely generated.

It follows that flips exist (but the termination of flips conjecture is open).

Higher dimensional boundedness

One can also generalize some of the features of the classification of surfaces. For example:

Theorem (Tsuji, Hacon-M$^{\mathrm{c}}$Kernan, Takayama)

For any integers $d, M > 0$, there are only finitely many families of varieties of general type, dimension d and volume $\mathrm{Vol}(\omega_X) = c_1(\omega_{X_{\min}})^d \leq M$, modulo birational isomorphisms.

In particular the set $V_d = \{\mathrm{Vol}(\omega_X) \mid \dim X = d\}$ is discrete.

Moduli spaces of canonical polarized varieties

Using techniques of Viehweg, Kollár, Shephered-Barron, Alexeev and others, one can refine the above results to show:

Theorem

Fix d and M. Then the set of canonical models X_{can} of varieties with $\dim X = \kappa(X) = d$ and $\mathrm{Vol}(\omega_X) \le M$ admits a coarse moduli space $\mathcal{M}_{d,M}$. In particular, the canonical models X_{can} as above are in 1-1 correspondence with the points of $\mathcal{M}_{d,M}$.

Naturally, one would like to compactify $\mathcal{M}_{d,M}$ in a geometrically meaningful way. In particular one should understand the natural degenerations of canonical models.

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- Let $\mathcal{X} = \mathrm{Proj}_\Delta R(\omega_{\mathcal{X}'})$. Then $\mathcal{X}$ agrees with $\mathcal{X}^*$ over Δ^* and $\mathcal{X}_0 = Y_1 \cup \ldots \cup Y_n$ is SLC (i.e. has mild singularities, e.g. a union of smooth codimension 1 subvarieties meeting transversely).

Degenerations of canonical models

- Given a family $\mathcal{X}^* \to \Delta^* = \Delta - \{0\}$ of canonical models, consider a compactification $\overline{\mathcal{X}} \to \Delta$.
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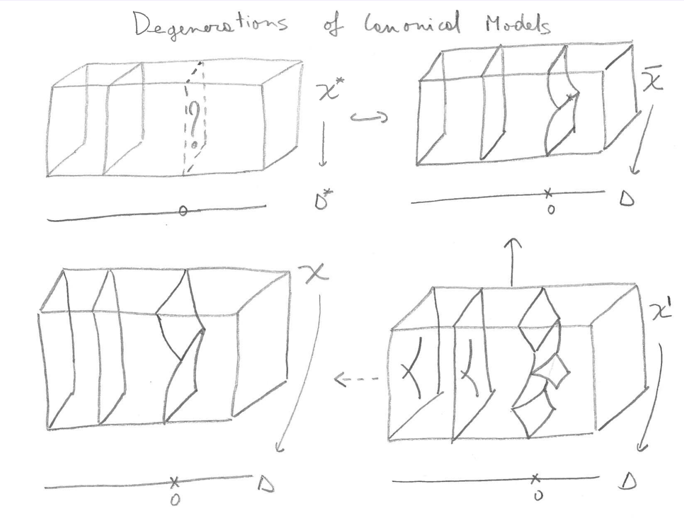
- This procedure gives us a geometrically meaningful compactification $\mathcal{M}_{d,M} \subset \overline{\mathcal{M}_{d,M}}$.

Degenerations



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- In particular we allow poles along $Y \cap Y_j \subset Y$ (and other singularities).
- New technical issues and difficulties arise when studying the geometry of these pairs (Y, B). However there have been sveral recent breakthroughs. For example we have the following:

Moduli spaces of canonical models

Theorem (Hacon-McKernan-Xu)

For any integer $d > 0$ the **set of volumes** $\mathrm{Vol}(\omega_X(\sum(1-\frac{1}{n_i}B_i)))$ *where X is smooth,* $\dim X = d$, *$\sum B_i$ is a union of codimension 1 subvarieties, and $n_i \in \mathbb{N}$* **has no accumulation points from above**. *In particular there exists $\epsilon > 0$ such that if* $\mathrm{Vol}(\omega_X(\sum(1-\frac{1}{n_i}B_i))) > 0$, *then* $\mathrm{Vol}(\omega_X(\sum(1-\frac{1}{n_i}B_i))) > \epsilon$. *Eg. if* $\dim X = 1$ *then* $\epsilon = 1/42$.

It appears that all the pieces are now in place to construct the moduli space of canonical models of SLC pairs (X, B) (thanks to work of Kollár, Abramovich, Alexeev, Hacking, Hacon, Hassett, Karu, Kovács, McKernan, Viehweg, Xu and others).